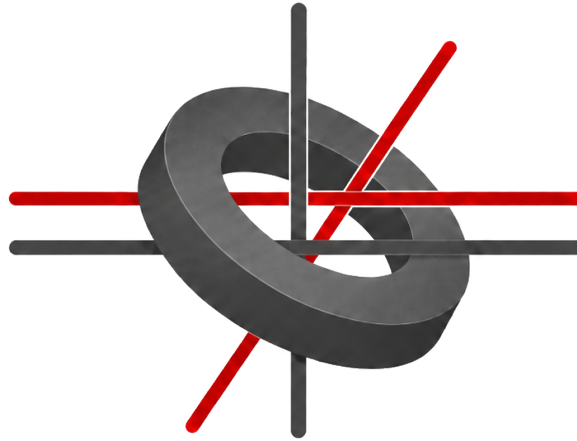




Final Degree Project Documentation in Computer Engineering in the AI Era

An unofficial, academic and practical guide for TFG documentation at USAL

Alvaro Lozano Murciego

CONTENTS

I	TFG Regulations	3
1	TFG Regulations	5
1.1	Course Description	5
1.2	What Is a TFG in Computer Engineering?	6
1.3	USAL Regulations	6
1.4	Faculty of Sciences Regulations	7
1.5	Department Regulations	7
1.6	Relationship with Professional Standards	9
II	TFG Documentation	11
2	TFG Documentation	13
2.1	Style Rules	13
2.2	Citations and Bibliography	13
2.3	Current Structure	16
2.4	Old ITIS/PFC Structure	16
2.5	Old vs Current Mapping	16
3	The Report	17
3.1	What should accompany the report	17
3.2	How to use the sections	17
3.3	Introduction	18
3.4	Purpose and objectives	18
3.5	Background and current situation	19
3.6	Methods, tools, and models	20
3.7	Requirements, scope, and constraints	21
3.8	Proposed solution	22
3.9	Risks and management	23
3.10	Conclusions	24
4	Annexes	27
4.1	A1. System specifications	27
4.2	A2. System analysis and design	27
4.3	A3. Size and effort estimation	27
4.4	A4. Security plan	27
4.5	A5. Other annexes	27

III	Software Engineering	29
5	Software Engineering Applied to the TFG	31
5.1	RUP vs Agile	31
5.2	UML and Documentation as Code	31
5.3	Planning, Risks and Tests	31
IV	Tools	33
6	Documentation Tools	35
6.1	Team Word	35
6.2	Team LaTeX	35
6.3	VS Code, AI Agents and Skills	36
V	Mistakes and FAQ	37
7	Common Mistakes and FAQ	39
7.1	Common Mistakes	39
7.2	FAQ	39
VI	Defense	41
8	TFG Presentation and Defense	43
VII	Final Information	45
9	Credits	47
10	Licenses	49
11	How to Cite	51
12	Bibliography	53

This English version is a structural placeholder. The first complete version of the book is maintained in Spanish.

Note

Translation pending. Always use the Spanish version as the source of truth for now.

The book is an unofficial, opinionated and practical guide for documenting a Final Degree Project in Computer Engineering at the University of Salamanca.

Part I

TFG Regulations

TFG REGULATIONS

The first practical decision when starting the documentation for the Final Degree Project (TFG) in Computer Engineering is to review the regulations that govern the subject. In the Computer Engineering degree there are three levels that should be kept separate:

- The general TFG regulations of the University of Salamanca.
- The regulations and procedures of the Faculty of Sciences.
- The specific page of the Department of Computer Science and Automation for the Computer Engineering degree.

In addition, the technical documentation recommended by the Department is related to professional standards for Computer Engineering projects. This relationship does not turn your TFG into a complete formally certified professional project, but it does explain the origin of this documentation and why the report, technical annexes, traceability, planning, and justification of decisions are required.

Tip

Always keep the official links up to date and record the version or academic year consulted. Web pages change; your report must be able to justify which documentation you followed.

1.1 Course Description

The academic course description in USAL Guides is another official source that should be located and reviewed in its current version. The description consulted identifies the Final Degree Project for the Computer Engineering degree with code 101147, 12 ECTS, Final Degree Project status, and placement in the 4th year ([Universidad de Salamanca, 2025](#)).

This description does not replace the regulations or the instructions issued by the Faculty and the Department. Its value lies in defining the academic nature of the subject: it is not an extended practical assignment or a simple software demo, but a final subject with a report, supervision through tutorials (with assigned hours), and a public defense before an evaluation panel.

1.2 What Is a TFG in Computer Engineering?

In the Computer Engineering degree, the TFG is an **integrative** piece of work: it must bring together degree competences and apply them to a project or piece of work related to the contents of the degree. The official course description presents it as a synthesis exercise in which **the student learns to develop, document, present, and properly defend their work before a specialized audience** (Universidad de Salamanca, 2025).

In practical terms, this implies three responsibilities:

- **Build something with sufficient technical substance:** software, a prototype, applied research, analysis, integration, a tool, a system, or a study related to computer science.
- **Document it** so that another person can understand the problem, the decisions, the solution, the tests, the limitations, and future work.
- **Defend it orally** with technical rigor, clarity, and command of the subject.

That is why this book insists that documentation is not later bureaucracy. It is part of the TFG itself: it records evidence of how you worked and allows the panel to evaluate not only the final result, but also the engineering judgment behind the decisions made. In an academic world with AI, it is more important than ever to show that the work is yours, that you understand what you have done, and that you can explain it to others.

1.3 USAL Regulations

The University of Salamanca regulations define the common framework for all Final Degree Projects: the nature of the TFG, enrollment, organization, assessment, and public defense. The Faculty and the Department adapt that framework to each degree.

For a Computer Engineering student, the important point is not to read the general regulations as an isolated document. They help explain shared principles:

- The TFG is a compulsory subject in the curriculum.
- The defense takes place in a public session.
- Assessment follows the regulations of the center and the degree.
- The student must meet the academic requirements before defending the project.

The Faculty page links to the University of Salamanca Final Degree Project Regulations, approved by the Governing Council on May 27, 2022 and amended on March 24, 2023 (Universidad de Salamanca, 2023).

1.3.1 What Should the Student Do?

1. Check the regulations in force for the corresponding academic year.
2. Review whether there are changes in the calendar, submission, or defense process.
3. Confirm any administrative requirement questions with the supervisor or TFG coordinator.
4. Do not copy regulatory fragments into the report unless they add something specific; complying with them is usually enough.

The report must demonstrate the work carried out, not become a compilation of regulations.

1.4 Faculty of Sciences Regulations

The TFG page of the Faculty of Sciences is the central point for regulations, scheduling, forms, common templates, and degree-specific links ([Facultad de Ciencias, Universidad de Salamanca, 2026](#)).

That page includes, among other resources:

- The University of Salamanca TFG regulations.
- The Faculty of Sciences regulations for TFG management.
- The academic-year schedule.
- TFG committees.
- Common templates and forms.
- Specific information for the Computer Engineering degree.

1.4.1 Schedule

The annual schedule is critical because it sets the deadlines for enrollment, proposal, assignment, submission, supervisor report, defense, and closing of minutes. Each year, the Faculty publishes a schedule document that distinguishes general actions from specific deadlines for the Computer Engineering degree ([Facultad de Ciencias, Universidad de Salamanca, 2025](#)).

Warning

Do not use dates from previous academic years as an operational reference. This book can explain the process, but the real dates must be checked every year.

1.4.2 Submission and Defense

The Faculty states that submission of the report and the presentation document is carried out through Studium within the established dates. From there, the committee assigns the evaluation panel or committee as applicable.

For documentation, this has a direct consequence: the report and the presentation are not prepared at the end. **They are prepared with enough time for the supervisor to review them and for the student to make corrections before submission.** There are usually several rounds of review and correction. Based on experience from previous years, the review and correction process **usually takes approximately one month.** Do not start preparing the documentation at the last minute, because it is very likely that there will not be enough time to correct it and prepare the final version for submission.

1.5 Department Regulations

The Department of Computer Science and Automation page (Diaweb) for the TFG in the Computer Engineering degree is the main living source for this degree ([Departamento de Informática y Automática, Universidad de Salamanca, 2026](#)).

It explains practical aspects such as:

- The nature of the TFG as an integrative exercise.
- Enrollment and requirements for defense.

- Choosing or proposing a topic.
- Approval of proposals by the TFG Committee.
- Preparation and submission of documentation.
- Defense, presentation times, and questions.
- General grading criteria.
- Templates and useful links.

1.5.1 Documentation

Diaweb states that, in order to defend the TFG, the student must submit the documentation and the artifacts resulting from the work: source code, binaries, libraries, datasets, or other relevant materials.

Regarding the report, it links to the documentation guide adapted for Computer Engineering TFGs. That guide takes as its reference an academic subset of the technical project standards officially disseminated by CPIICM and adapts it to the needs of the degree ([Colegio Profesional de Ingenieros en Informática de la Comunidad de Madrid, 2016](#), [Colegio Profesional de Ingenieros en Informática de la Comunidad de Madrid, 2026](#), [Comisión de Trabajos Fin de Grado del Grado en Ingeniería Informática, 2024](#)).

Important

The adapted guide should not be read as a blind template. It must be adapted to the nature of the TFG, and the omission of documents or sections that are not relevant must be justified.

1.5.2 Assessment

The Department summarizes general assessment criteria: quality of the report and annexes, complexity and command of the topic, oral and written presentation, quality of prototypes when applicable, supervisor report, and other criteria considered by the panel.

The Faculty regulations include a rubric that the panel must use to evaluate each part of the report, the presentation, and the work as a whole. This rubric is based on criteria such as technical quality, clarity, rigor, originality, justification of decisions, and other aspects related to Computer Engineering ([Facultad de Ciencias, Universidad de Salamanca, 2022](#)).

The rubric can be summarized as follows:

As-sessed area	Weight in the grade	Assessment criteria	Minimum to average
Report assessment	25%	Complete, coherent, and detailed statement of objectives, presentation of results, and conclusions; correct description of the technical part, including annexes, design, methodology, and assessments; correctness of format and structure, including tables, diagrams, figures, bibliography, and writing.	1.0
Oral defense assessment	15%	Presentation and oral communication; clear, understandable, and ordered explanation of ideas and concepts; discussion and defense of the work.	0.6
Technical assessment of the TFG	60%	Final quality of the work; degree of difficulty and alignment with the proposed objectives; correct use of methodologies and techniques.	2.4

The minimum required to average is 40% in each area. The sum of those minimums is 4.0 points out of the final grade.

1.6 Relationship with Professional Standards

The Department page recommends following a model adapted from professional standards for Computer Engineering projects ([Departamento de Informática y Automática, Universidad de Salamanca, 2026](#)).

The official source for this professional context is the CPIICM page on Computer Engineering project standards. It presents the Technical Standard for Computer Engineering Projects prepared by the Professional Association of Computer Engineers of the Community of Madrid and the Official Association of Computer Engineers of the Basque Country ([Colegio Profesional de Ingenieros en Informática de la Comunidad de Madrid, 2026](#)).

Specifically, the current Department documentation for the TFG should be understood as an academic subset adapted from those professional standards. For the documentary details, this book uses as a technical reference the CPIICM Technical Standard PDF for preparing Computer Engineering project documentation ([Colegio Profesional de Ingenieros en Informática de la Comunidad de Madrid, 2016](#)), in relation to Standard CCII-N2016-02 for preparing Computer Engineering project documentation ([Consejo General de Colegios Profesionales de Ingeniería en Informática, 2016](#)).

1.6.1 What Does This Mean for a TFG?

It does not mean that the TFG is a complete formally certified professional project or that every document from a certified project must be reproduced. It means that the TFG guide selects and adapts the elements that make academic sense:

- The report connects and justifies the whole project.
- The annexes develop the technical part in greater detail.
- Relevant decisions must be traceable.
- Requirements, analysis, design, planning, testing, and security are not documentary decoration.

Part II

TFG Documentation

TFG DOCUMENTATION

Translation pending. Use the Spanish version as the source of truth.

2.1 Style Rules

Translation pending. Use the Spanish version as the source of truth.

2.2 Citations and Bibliography

In this guide the criterion is deliberately strict: **a serious TFG should use a bibliography manager**. Mendeley and Zotero are reasonable options; choosing one or the other is less important than not managing references by hand.

Managing citations manually usually leads to errors:

- References that appear in the bibliography but are not cited.
- In-text citations that do not appear in the bibliography.
- Incomplete dates, authors or titles.
- Citation styles applied inconsistently.
- URLs without access dates.
- Bibliographies that differ between Word, PDF and LaTeX.

2.2.1 Minimum Recommended Workflow

1. Install a bibliography manager.
2. Capture sources from the browser.
3. Review the metadata before accepting the reference.
4. Organize sources in folders or collections for the TFG.
5. Insert citations from the Word plugin or export BibTeX for LaTeX/Overleaf.
6. Generate the bibliography automatically.
7. Check at the end that every citation has an entry and every entry is cited.

2.2.2 Mendeley As An Example

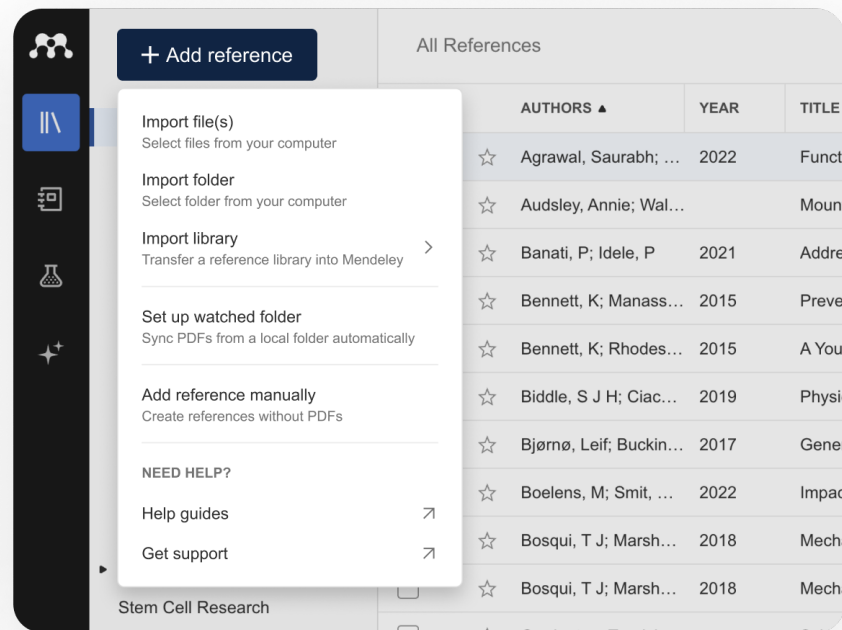


Figure2.1: Mendeley Reference Manager can import PDFs and organize references in a library.

Image source: Mendeley (Elsevier, 2026).

A comfortable workflow with Mendeley is:

- Use Mendeley Reference Manager as the central library (Elsevier, 2026).
- Install Mendeley Web Importer in Chrome or a compatible browser to capture articles, pages and PDFs (Elsevier, 2026).
- Review authors, title, year, DOI, URL and access date.
- In Word, insert citations with the Mendeley add-in.
- In LaTeX or Overleaf, export a collection to BibTeX or keep a synchronized .bib file.

The critical point is not clicking “import”. The critical point is reviewing. Many importers confuse capitalization, web pages, conference names or corporate authors.

2.2.3 Zotero As An Alternative

Zotero is a very solid alternative, especially if you want an open ecosystem, good browser capture and BibTeX/BibLaTeX export (Corporation for Digital Scholarship, 2026).

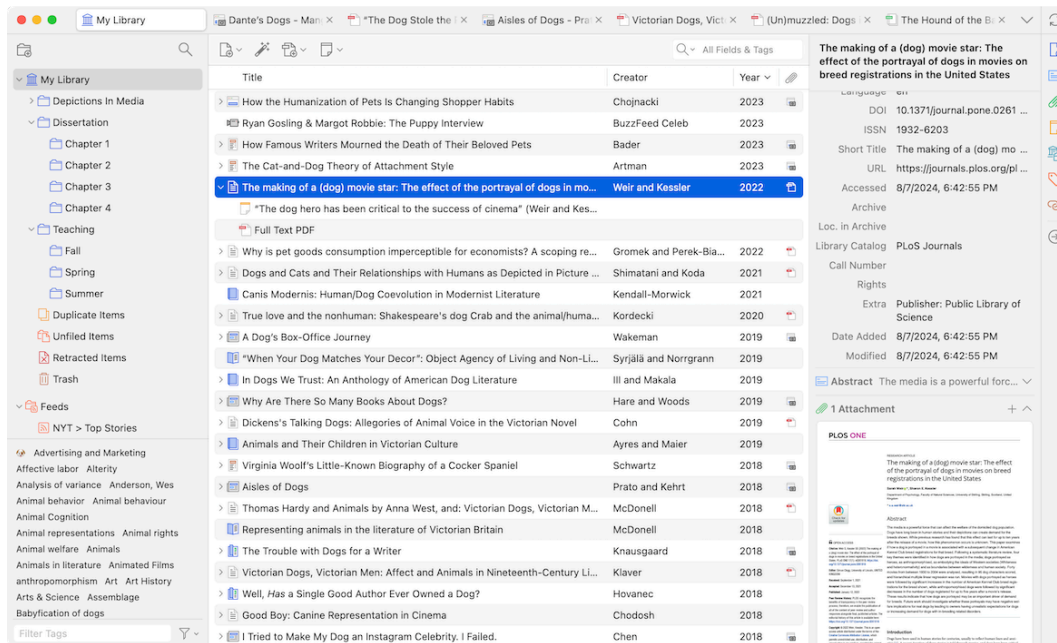


Figure 2.2: Zotero organizes sources by collections and shows metadata, attachments and notes in the same interface.

Image source: Zotero (Corporation for Digital Scholarship, 2026).

2.2.4 Bibliographic Style

If your supervisor, committee or current regulations require a specific style, use that. The DIA documentation guide states that the bibliography should be cited properly following whichever standard is considered most appropriate (Comisión de Trabajos Fin de Grado del Grado en Ingeniería Informática, 2024).

This book uses an **APA-inspired author-year system**. A parenthetical citation should appear as (*Author*, *year*) and a narrative citation can be integrated as *Author* (*year*). The final bibliography is displayed without visible numbering and with a hanging indent, so it is closer to the usual appearance of an APA reference list.

APA In This Book

The book configuration forces author-year citations with parentheses and hides numeric bibliography labels in the final web list. Even so, it is better to describe this as *APA-like* or *author-year* style, not as strict APA 7th edition compliance, because the list is generated by Jupyter Book/Sphinx from BibTeX.

In a TFG report, author-year citations make it easier to see where each idea comes from without forcing the reader to jump continuously to a numbered list. If your supervisor, committee or an official template requires IEEE or another style, that requirement takes precedence. In any case, consistency matters more than personal preference.

Important

AI can help you detect incomplete references, but it must not invent sources. Every reference must exist, be relevant and have been consulted by the author.

2.3 Current Structure

Translation pending. Use the Spanish version as the source of truth.

2.4 Old ITIS/PFC Structure

Translation pending. Use the Spanish version as the source of truth.

2.5 Old vs Current Mapping

Translation pending. Use the Spanish version as the source of truth.

THE REPORT

The report is the document that the assessment committee should be able to read from start to finish in order to understand the TFG. It should not contain every technical detail, but it must contain the important decisions: what problem is addressed, why it matters, what has been built or studied, how it has been validated and what can be concluded.

The DIA v2 guide presents the report as the document that connects the rest of the TFG documentation and justifies the adopted solutions (Comisión de Trabajos Fin de Grado del Grado en Ingeniería Informática, 2024). That is the central idea: the report does not replace the annexes; it guides them. The reader should be able to follow the main reasoning without reading every detailed requirement, diagram or test log.

3.1 What should accompany the report

Before the technical chapters, the documentation should include several formal elements:

- **Cover page:** clear title, author, defense date and supervisor or supervisors. The title should be meaningful and avoid unexplained acronyms.
- **Supervisor consent form:** include the current official model when the call or Department requires it.
- **Resumen and abstract:** self-contained synthesis in Spanish and English, usually between 200 and 500 words. It should include the problem, methodology, tools, results and conclusions.
- **Glossary:** useful when the report uses domain terms, non-standard abbreviations or acronyms.
- **Indexes:** report table of contents, annex index and tables/figures indexes when used.

Figures and tables need captions and textual references. External resources must be cited. Mandatory requirements should use clear obligation wording: “must”, “shall” or equivalent. Optional improvements should be marked as desirable rather than compulsory.

3.2 How to use the sections

The following sections should remain connected:

- Objectives explain what the TFG tries to achieve.
- Background justifies why the problem deserves attention.
- Methods, tools and models explain the approach.
- Requirements, constraints and scope define the limits.
- The proposed solution shows what has been produced.
- Management, testing and risks show control of the process.

- Conclusions return to the objectives and assess the result.

When an explanation becomes too detailed for the report, move it to an annex and leave a summary and cross-reference here.

3.3 Introduction

The introduction opens the report and should orient the committee before entering technical detail. According to the DIA v2 guide, it should present the problem domain, its context, possible solutions, the motivation for the work and the overall structure of the documentation ([Comisión de Trabajos Fin de Grado del Grado en Ingeniería Informática, 2024](#)).

A useful introduction answers these questions early:

- What domain does the TFG address?
- What problem, need or opportunity has been identified?
- Why does it make sense as a Computer Engineering TFG?
- What general approach has been followed?
- How is the report organized?

Do not duplicate the abstract. The abstract condenses the whole work; the introduction prepares the reader. It may mention general alternatives, but it should not reveal the whole design or become a tutorial on the technologies used.

3.3.1 Recommended structure

A solid structure is:

1. Present the context and application area.
2. Explain the problem with concrete examples.
3. Justify the academic, technical or social motivation.
4. Anticipate the type of solution or study performed.
5. Close with a short map of the document.

Keep the wording precise. Avoid generic statements if they do not support the project. It is better to explain what real limitation exists, who it affects and why your work provides an assessable response.

3.4 Purpose and objectives

This section defines what the TFG aims to achieve. The DIA v2 guide asks for functional and personal objectives, with one main objective and several achievable, reasonable and measurable subobjectives ([Comisión de Trabajos Fin de Grado del Grado en Ingeniería Informática, 2024](#)).

The **main objective** should express the central contribution: building a system, designing a method, evaluating a technique, developing a prototype, automating a process, comparing alternatives or generating applicable knowledge.

The **subobjectives** split that promise into verifiable results. They are often clearer when written with action verbs:

- Analyze the problem and its constraints.
- Design an architecture or solution model.
- Implement a working prototype.

- Evaluate the solution through metrics, tests or user validation.
- Document the process so that it is reproducible.

3.4.1 Functional and personal objectives

Functional objectives describe what the work should produce. Personal objectives describe relevant learning by the author, but only when connected to the TFG. “Learning a technology” is not enough; it is better to state that a specific technology will be applied and evaluated to solve a defined need.

3.4.2 Quality criteria

An objective is well written if the final report can answer:

- Has it been achieved?
- With what evidence?
- What limitation remains?
- Which technical decision helped achieve it?

Do not turn this section into a wish list. If a subobjective cannot be verified in the report, annexes, repository or tests, it probably needs rewriting.

3.5 Background and current situation

This block has two functions: explaining the background of the problem and describing the current situation. The DIA v2 guide separates both points: background gives the theoretical or empirical basis of the domain, while the current situation presents a state of the art, market study or existing solutions depending on the TFG ([Comisión de Trabajos Fin de Grado del Grado en Ingeniería Informática, 2024](#)).

3.5.1 Background

The background should explain the non-computing domain when needed. If the TFG deals with education, health, business management, heritage, sports or scientific research, this section should give the committee the minimum basis needed to understand the problem.

Include:

- essential domain concepts;
- relevant previous studies or works;
- limitations in previous solutions;
- needs of users, organizations or processes;
- company context, only when applicable and allowed.

Theory should serve the TFG. There is no need to write a manual for every technology used; explain the prior knowledge that the committee needs in order to understand your decisions.

3.5.2 Current situation

Compare existing alternatives: similar applications, reference tools, commercial products, repositories, papers, regulations or common practices. If technological evolution matters, describe it briefly, but avoid a long history with no connection to your solution.

This section should rely on references. Do not simply claim that no adequate solution exists. Show what has been reviewed, what each alternative offers and why your chosen path makes sense.

3.6 Methods, tools, and models

The DIA v2 guide groups here standards, references, methods, tools, models, prototypes and metrics ([Comisión de Trabajos Fin de Grado del Grado en Ingeniería Informática, 2024](#)). The goal is not to list everything installed, but to explain the technical approach and justify decisions that affect the result.

3.6.1 Methods

Describe the technical methodology: unified process, agile development, iterative prototyping, experimental methodology, systematic review, user-centered design or another approach. If you rely on known concepts, cite them and summarize only what is needed.

The method should explain how the work moved from problem to solution:

- project phases;
- decision criteria;
- validation strategy;
- relation between methodology and planning.

3.6.2 Tools

Group tools by function:

- **Implementation:** languages, frameworks, libraries, databases, APIs, cloud services, devices or models.
- **Methodology:** task management, version control, continuous integration, testing, design, prototyping or technical documentation.
- **TFG writing:** LaTeX, Markdown, Jupyter Book, bibliography managers, diagram tools or AI assistants when they materially influenced the work.

Do not explain standard tools if they add no value. Saying that a text editor was used is rarely relevant; explaining why a specific AI library, architecture or database was chosen is relevant.

3.6.3 Models, prototypes and metrics

Include models when they are part of the work: AI, mathematical, industrial, electronic, simulation, data or decision models. If there are wireframes, mock-ups or navigation diagrams, they should justify design and user experience.

Metrics are useful when they help assess the result: performance, accuracy, test coverage, user satisfaction, execution time, resource use, accessibility or documentation quality. A metric without interpretation proves little; always pair it with a criterion.

3.7 Requirements, scope, and constraints

The reader should understand what is inside and outside the TFG. The DIA v2 guide recommends summarizing initial requirements, hypotheses, constraints, scope, expected impact and the alternatives and feasibility study in the report (Comisión de Trabajos Fin de Grado del Grado en Ingeniería Informática, 2024).

3.7.1 Initial requirements

The report does not need to repeat the whole A1 annex. It should present a summary that makes the commitment clear:

- main functional requirements;
- relevant non-functional requirements;
- use cases, user stories or equivalent format;
- acceptance criteria;
- data, security, performance, accessibility, deployment or maintenance constraints.

Mandatory requirements should be written as commitments: “the system must...”, “the application shall...”. Use “should” for desirable improvements outside the minimum scope.

3.7.2 Hypotheses, constraints and scope

Constraints are not excuses; they are design conditions. They may come from available time, data, regulations, privacy, hardware, budget, APIs, prior knowledge, integration needs or limited access to users.

The scope should make clear:

- what the TFG includes;
- what is outside the work;
- what is left as future work;
- what level of completion is expected: prototype, usable product, research work, library, demonstrator or technical documentation.

It is also useful to explain the expected impact: which user, organization, process or sector benefits from the solution and how.

3.7.3 Alternatives and feasibility

The report should show that technologies were not chosen randomly. Compare alternatives for languages, frameworks, APIs, databases, models or services. The comparison can be brief, but it should justify the choice using criteria such as cost, maturity, documentation, performance, privacy, deployment, community, license or compatibility with requirements.

When applicable, add a qualitative assessment of costs and benefits. In a TFG, monetization is not always relevant, but effort, maintenance, reuse and real feasibility usually are.

3.8 Proposed solution

The proposed solution should technically explain the final product, method or result. The DIA v2 guide asks for functionalities, screenshots, diagrams and a way to test the development: deployment, executable, installer or demonstration video when applicable ([Comisión de Trabajos Fin de Grado del Grado en Ingeniería Informática, 2024](#)).

This section must connect requirements and outcome. It is not enough to show screens; explain the relevant technical decisions, main components and how the solution responds to the objectives.

3.8.1 What should be clear

Include, depending on the TFG:

- overview of the solution;
- system architecture;
- main components and responsibilities;
- data or usage flow;
- implemented functionalities;
- screenshots or diagrams with explanation;
- technical decisions and discarded alternatives;
- repository, demo, deployment, installer or video link;
- tests or evidence of operation.

Diagrams should be referenced in the text and have captions. A screenshot without explanation has little value; a screenshot showing how a requirement is solved does.

3.8.2 From proposal to result

Explain how the solution evolved: problems found, decisions made, scope changes, limitations and particularities of the final version. This often separates a mature report from a superficial description.

If the solution includes AI, external models or APIs, document what the author controls and what depends on third-party services. If data is used, explain its origin, treatment and limitations. If there is deployment, state how it can be reproduced or tested.

3.8.3 Validation evidence

Close the solution with evidence: tests, metrics, user validation, comparison with requirements or result analysis. If the detail is large, keep the synthesis here and move full tables, cases or logs to the relevant annex.

3.9 Risks and management

This block should show that the project was managed with judgment. The DIA v2 guide distinguishes risk analysis, organization and project management ([Comisión de Trabajos Fin de Grado del Grado en Ingeniería Informática, 2024](#)). The report should not include a decorative Gantt chart; it should explain how work was organized and how problems were handled.

3.9.1 Risk analysis

Select risks that truly affect the TFG:

- **Acceptance:** users do not understand, need or adopt the solution.
- **Technology:** a library, API, model, device or architecture does not behave as expected.
- **Data:** lack of data, bias, poor quality, privacy or permissions.
- **Integration:** problems connecting the solution to a larger system.
- **Planning:** underestimated tasks, external dependencies or scope changes.

You may use a risk matrix, SWOT analysis or risk register. A minimal register can include number, description, priority, complexity and corrective measures. What matters is that each risk has a response: accept, reduce, avoid, transfer or monitor.

3.9.2 Work organization

Explain the real phases of the project and how the chosen methodology was applied. The path from proposal to final solution should be visible:

- initial analysis;
- design;
- implementation;
- testing;
- documentation review;
- defense preparation.

When architecture, components, protocols, entities, databases or algorithms are important, summarize the organization here and refer to the analysis and design annex for detail.

3.9.3 Project management

Include resources, timeline, milestones and deviations. If changes occurred, explain them naturally: a serious TFG is not one without problems, but one that identifies, manages and justifies adjustments.

The Gantt chart, if included, should tell the real story of the work. Connect planning, risks, decisions and final scope.

3.10 Conclusions

The conclusions close the report by returning to the objectives. The DIA v2 guide asks the author to make clear whether the objectives were achieved, what problems appeared, how they were solved, what technical and personal learning took place, what limitations remain and what future lines of work are possible ([Comisión de Trabajos Fin de Grado del Grado en Ingeniería Informática, 2024](#)).

A serious conclusion does not merely state that the objectives were met. It should show it:

- objective;
- result obtained;
- evidence used to verify it;
- limitation found;
- possible improvement.

3.10.1 Results and learning

Summarize the main results without repeating the whole solution. Explain what worked, what did not go as far as expected and which decisions mattered most.

It is also reasonable to include professional and personal learning, as long as it is connected to the TFG: technologies learned, design judgment, time management, communication with users, data handling, documentation, testing or deployment.

3.10.2 Limitations and future work

Limitations are not failures; they are the honest boundary of the work. They may relate to time, data, performance, scale, validation, user access, privacy, generalization or maintenance.

Future work should be realistic and connected to what was actually done:

- extending a delimited functionality;
- improving performance or accessibility;
- validating with more users or data;
- integrating with another system;
- strengthening security or deployment;
- turning a prototype into a maintainable product.

3.10.3 Bibliography before submission

The DIA v2 guide recommends a standard citation format and mentions APA as a practical option ([Comisión de Trabajos Fin de Grado del Grado en Ingeniería Informática, 2024](#)). This book uses an APA-like author-year style. Before submission, check three things: every in-text citation appears in the bibliography, every bibliography entry has been cited at least once, and the sources are relevant to the TFG topic.

Translation pending. Use the Spanish version as the source of truth.

4.1 A1. System specifications

Translation pending. Use the Spanish version as the source of truth.

4.2 A2. System analysis and design

Translation pending. Use the Spanish version as the source of truth.

4.3 A3. Size and effort estimation

Translation pending. Use the Spanish version as the source of truth.

4.4 A4. Security plan

Translation pending. Use the Spanish version as the source of truth.

4.5 A5. Other annexes

Translation pending. Use the Spanish version as the source of truth.

Part III

Software Engineering

SOFTWARE ENGINEERING APPLIED TO THE TFG

Translation pending. Use the Spanish version as the source of truth.

5.1 RUP vs Agile

Translation pending. Use the Spanish version as the source of truth.

5.2 UML and Documentation as Code

Translation pending. Use the Spanish version as the source of truth.

5.3 Planning, Risks and Tests

Translation pending. Use the Spanish version as the source of truth.

Part IV

Tools

DOCUMENTATION TOOLS

Translation pending. Use the Spanish version as the source of truth.

6.1 Team Word

Translation pending. Use the Spanish version as the source of truth.

6.2 Team LaTeX

Translation pending. Use the Spanish version as the source of truth.

6.2.1 Starting templates in Overleaf

Public Overleaf templates can be useful as technical starting points for folder structure, LaTeX configuration, bibliography handling, and PDF layout ([Overleaf, 2026](#)). They are not official USAL or DIA templates. When using an external template, adapt it to the current TFG rules, review its license, remove institutional text that does not apply, and cite the original template.

Examples worth reviewing include the University of Seville TFG template ([Trinidad, 2026](#)), the ULL-ESIT-GII TFG report template ([Torres, 2026](#)), the University of Málaga TFG report template ([Hidalgo, 2026](#)), the University of Cádiz software-development TFG template ([Profesorado del Departamento de Ingeniería Informática de la Universidad de Cádiz, 2026](#)), and the UDIMA Computer Engineering TFG template ([Barrio, 2026](#)).

6.2.2 Overleaf from VS Code

If you want to write in VS Code while keeping the project in Overleaf, the cleanest route is usually Overleaf's Git integration, which lets you clone the project, work locally, and synchronize changes with `pull` and `push` ([Overleaf, 2026](#)). Another option is the community extension [Overleaf Workshop](#), which opens Overleaf/ShareLaTeX projects in VS Code and supports editing, compiling, previewing, and collaboration features ([iamhyc, 2026](#)).

Treat any extension-based workflow as an external tool: check authentication, permissions, project state, and conflicts before relying on it for final TFG work.

6.3 VS Code, AI Agents and Skills

Translation pending. Use the Spanish version as the source of truth.

Part V

Mistakes and FAQ

COMMON MISTAKES AND FAQ

Translation pending. Use the Spanish version as the source of truth.

7.1 Common Mistakes

Translation pending. Use the Spanish version as the source of truth.

7.2 FAQ

Translation pending. Use the Spanish version as the source of truth.

Part VI

Defense

TFG PRESENTATION AND DEFENSE

Translation pending. Use the Spanish version as the source of truth.

Part VII

Final Information

CREDITS

Author: **Alvaro Lozano Murciego**.

Publication year: **2026**.

This English section is a placeholder. The Spanish version is the source of truth for the first public version of the book.

LICENSES

Unless stated otherwise, the original text of this book is distributed under the [Creative Commons Attribution 4.0 International License \(CC BY 4.0\)](#).

External institutional documents, logos, templates and linked resources keep their own terms of use.

HOW TO CITE

Suggested citation:

Lozano Murciego, A. (2026). *Documentacion de un TFG en Ingenieria Informatica en tiempos de IA*. Universidad de Salamanca. CC BY 4.0.

```
@book{lozano2026tfgia,  
  author = {Lozano Murciego, Alvaro},  
  title = {Documentacion de un TFG en Ingenieria Informatica en tiempos de IA},  
  year = {2026},  
  publisher = {Universidad de Salamanca},  
  note = {Unofficial guide, CC BY 4.0}  
}
```


- [10] Corporation for Digital Scholarship. Zotero Documentation. 2026. URL: <https://www.zotero.org/support/> (visited on 2026-05-28).
- [11] Corporation for Digital Scholarship. Zotero. 2026. URL: <https://www.zotero.org/> (visited on 2026-06-03).
- [12] Departamento de Informática y Automática, Universidad de Salamanca. Trabajos/Proyectos Finales: Grado en Ingeniería Informática, curso 2025/2026. 2026. URL: https://diaweb.usal.es/diaweb/asignaturas/proyectos/presentacion.jsp?cod\TU\textbackslash{ }_titulacion=211\TU\textbackslash{ }\TU\textbackslash{ }&cod\TU\textbackslash{ }_curso\TU\textbackslash{ }_academico=2025 (visited on 2026-05-28).
- [13] Elsevier. Mendeley Features. 2026. URL: <https://www.mendeley.com/features/> (visited on 2026-06-03).
- [14] Elsevier. Mendeley Reference Manager. 2026. URL: <https://www.mendeley.com/reference-management/reference-manager> (visited on 2026-05-28).
- [15] Elsevier. Mendeley Web Importer. 2026. URL: <https://www.mendeley.com/reference-management/web-importer> (visited on 2026-06-03).
- [16] Facultad de Ciencias, Universidad de Salamanca. Rúbrica (TFGII), versión 1.2. 2022. Rúbrica de evaluación para el Trabajo Fin de Grado del Grado en Ingeniería Informática. URL: <https://fciencias.usal.es/wp-content/uploads/2022/11/rubricagii-v1-2.pdf> (visited on 2026-06-03).
- [17] Facultad de Ciencias, Universidad de Salamanca. Trabajos Fin de Grado: Temporización Curso 2025-2026. 2025. URL: <https://fciencias.usal.es/wp-content/uploads/2025/10/modif.temporizacion-2025-2026.pdf> (visited on 2026-05-28).
- [18] Facultad de Ciencias, Universidad de Salamanca. Trabajos de Fin de Grado. 2026. URL: <https://fciencias.usal.es/grados/trabajos-de-fin-de-grado/> (visited on 2026-05-28).
- [19] Overleaf. Git integration. 2026. Documentación oficial de Overleaf sobre integración Git. URL: <https://docs.overleaf.com/integrations-and-add-ons/git-integration-and-github-synchronization/git-integration> (visited on 2026-05-28).
- [20] Overleaf. LaTeX templates: búsqueda de plantillas TFG ingeniería informática. 2026. URL: <https://www.overleaf.com/latex/templates?q=TFG+ingenieria+inform\TU\textbackslash{ }%C3\TU\textbackslash{ }%A1tica> (visited on 2026-05-28).
- [21] Profesorado del Departamento de Ingeniería Informática de la Universidad de Cádiz. Plantilla TFG GII UCA - desarrollo de software. 2026. Plantilla para TFG de desarrollo de software del Grado en Ingeniería Informática de la Universidad de Cádiz, publicada en Overleaf con licencia Creative Commons CC BY 4.0. URL: <https://es.overleaf.com/latex/templates/plantilla-tfg-gii-uca-desarrollo-de-software/zrbmfzbxkys> (visited on 2026-05-28).
- [22] Universidad de Salamanca. Reglamento de Trabajos de Fin de Grado de la Universidad de Salamanca. 2023. Aprobado el 27 de mayo de 2022 y modificado el 24 de marzo de 2023. URL: <https://fciencias.usal.es/grados/trabajos-de-fin-de-grado/> (visited on 2026-05-28).
- [23] Universidad de Salamanca. Guía académica: Trabajo Fin de Grado, Grado en Ingeniería Informática, curso 2025/2026. 2025. URL: <https://guias.usal.es/node/227026> (visited on 2026-05-28).